

Cleansing Oil^{*}

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Abstract

I study the dynamics of greenhouse gas emissions following disruptions in crude oil markets. Following a 10% unexpected increase in crude oil prices, CO₂-equivalent atmospheric concentration declines by 0.10 ppm, equivalent to 3.24% of its average yearly increase for the 21st century. The decline is heterogeneous across greenhouse gases and reflects a reshuffling of fossil fuel use rather than a transition toward cleaner energy. Natural gas and petroleum products decline substantially, while lignite production increases temporarily and bioenergy generation rises. Consistent with tighter financial conditions and elevated uncertainty, solar generation also contracts following the shock. I find no evidence of accelerated adoption of alternative-fuel vehicles. Overall, the results show that an oil shock provides some of the incentives for decarbonization associated with a carbon tax, but also generates distortions that a carbon tax does not.

1 Introduction

Oil supply disruptions are widely believed to accelerate the green transition and slow down global warming. Whenever geopolitical tensions threaten global crude oil supplies, it is routinely argued that higher fossil fuel prices will encourage the adoption of cleaner technologies and reduce greenhouse gas emissions. This paper shows that this view is only partially true.

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I find that a 10% increase in oil prices caused by an exogenous oil supply shock reduces global greenhouse-gas concentrations and increases emissions efficiency. However, the adjustment occurs mostly because of a decrease in energy consumption, and less from the substitution away from dirty and into green energy. Oil supply disruptions reduce emissions, but they do not accelerate the clean-energy transition, neither in the private vehicle market, nor in the power sector, at least in the short to medium run.

Oil supply disruptions trigger a broad reallocation across fuels rather than a simple substitution away from fossil energy. Consumption of petroleum products falls sharply, especially jet fuel, gasoline, and diesel. Natural gas consumption declines as well. At the same time, lignite production rises on impact and bioenergy generation increases. These adjustments generate heterogeneous greenhouse-gas dynamics. Atmospheric concentrations of carbon dioxide and methane decline, whereas nitrous oxide concentrations rise persistently. The reduction in overall warming potential therefore masks important changes in the composition of emissions and in the underlying energy mix.

The contraction in renewable deployment is particularly surprising. Conventional reasoning suggests that higher fossil-fuel prices should improve the competitiveness of clean alternatives and stimulate their adoption. Indeed, much of the discussion surrounding oil disruptions implicitly assumes that expensive oil acts as a catalyst for the green transition. Yet oil supply shocks are not carbon taxes. The same disturbance that raises the price of fossil fuels also depresses economic activity, increases uncertainty, tightens financial conditions, and disrupts complementary energy markets. These forces work against renewable investment and can dominate the standard relative-price channel, which is what I find.

The distinction matters for how oil disruptions should be viewed through a climate lens. An oil shock delivers a partial carbon-price signal, but it also generates distortions that a carbon tax does not. The resulting environmental consequences are therefore more nuanced than commonly assumed. Oil-intensive activities contract, yet renewable deployment weakens and electricity generation shifts toward lower-quality combustion alternatives such as lignite and bioenergy. Moreover, electricity reliance — as a secondary energy source — does not increase.

To study these effects, I bring the empirical tools of modern macroeconomics to the study of climate change and energy transitions. Combining identified oil supply shocks with global monthly data on greenhouse-gas concentrations, fossil-fuel consumption, electricity generation, renewable capacity, and vehicle registrations, I estimate the dynamic responses of emissions and energy use to exogenous disruptions in oil markets. The analysis spans the period from 2001 to 2025 and traces how oil supply shocks propagate through the energy system — from fuel consumption and electricity generation to technology adoption, and ultimately global greenhouse-gas concentrations.

1.1 Related Literature

This paper contributes to five strands of literature. First, it relates to the large literature studying the macroeconomic consequences and transmission channels of oil-market shocks. Beginning with Hamilton (1983), a vast literature has documented that oil-price increases are associated with declines in economic activity and changes in inflation, investment, and financial conditions. More recent contributions have emphasized the importance of distinguishing between different sources of oil-price fluctuations. For example, Kilian (2009) shows that supply, aggregate-demand, and precautionary-demand shocks have markedly different macroeconomic consequences, while Baumeister and Hamilton (2019) develop a Bayesian framework to incorporate prior knowledge about the different disturbances. Building on techniques typically used in the analysis of monetary policy, Känzig (2021) uses high-frequency information around OPEC announcements to identify exogenous oil-supply news shocks and study their macroeconomic effects. I leverage this literature for identification, but shift the focus towards the environmental and energy consequences of oil shocks.

Second, the paper contributes to the literature studying the response of fossil-fuel demand to energy price changes. Much of the attention in this literature has been devoted to gasoline markets — in particular, to estimating the price elasticity of gasoline demand. A growing body of work documents how this elasticity has evolved over time and varies across households and regions, with Kilian and Zhou (2024) providing a recent account using cross-state variation in oil-to-gasoline pass-through as a new instrument. Far less attention has been devoted to other petroleum derivatives and primary fuels: diesel, jet fuel, liquefied petroleum gas (LPG), natural gas, and coal each face distinct demand structures and substitution margins that a gasoline-centric lens cannot capture. The importance of this broader coverage is underscored by recent events: the threatened closure of the Strait of Hormuz in 2026 raised acute concerns about the supply of jet fuel for global aviation and bunker oil for international maritime trade. I contribute by documenting the response of the full fossil-fuel spectrum — gasoline, diesel, jet fuel, liquefied petroleum gas, natural gas, and coal — to exogenous oil-supply shocks.

Third, this paper relates to the literature studying the relationship between greenhouse-gas emissions and macroeconomic fluctuations. Heutel (2012) establishes that carbon emissions are deeply procyclical in the United States. Using a cross-section of countries and annual data, Doda (2014) confirms this procyclicality internationally and further documents that emissions are more volatile than output. Closer to my work, Khan et al. (2019) provide a first attempt at identifying the causal effects of different macroeconomic shocks — including oil-price and technology shocks — on US emissions, but do not find statistically significant effects for any of them. I document significant effects of oil-supply shocks on emissions, with the added value of conducting a world-wide analysis — important for a global problem like climate change.

Fourth, the paper contributes to the macroeconomic literature on energy efficiency, directed technical change, and capital structures. In their foundational work, Hassler, Krusell, and Olovsson (2021) show that the short-run substitution elasticity between energy and capital/labor is virtually zero (near-Leontief), but that sustained energy price increases direct innovation toward energy-saving technologies, rendering long-run substitutability significantly higher. Using time-series methods, Känzig and Williamson (2024) isolate these technological drivers, finding that while energy price shocks successfully stimulate energy-saving innovation, they are highly recessionary and crowd out other productive innovations. Conversely, pure energy-saving technology shocks drive a non-recessionary decoupling of output and emissions. Furthermore, Casey (2024) studies optimal taxation in a model with energy use and directed technical change. He argues that a Cobb-Douglas production function drastically overstates the long-run emissions reductions of simple energy taxes due to rebound effects that discourage subsequent innovation on energy-use, emphasizing the need for targeted R&D policy. I complement this literature by studying whether oil-price shocks improve emissions efficiency at the global level and by tracing the underlying mechanisms through detailed adjustments in fossil-fuel use, transportation demand, and electricity generation. I show that while an oil-price shock improves emissions efficiency per unit of industrial production, this is a short-lived effect, driven mostly by adjustments in the intensive margin. It does not translate into sustained emissions efficiency gains, nor does it spur an energy transition towards renewable energy sources.

Fifth, this paper contributes to the literature studying the adoption of clean-energy technologies and the structural and financial frictions that govern the green transition. While conventional theory suggests that fossil-fuel price spikes accelerate renewable adoption, macro-financial and technological frictions can disrupt this channel. First, renewable projects are highly capital-intensive and face upfront expenditures, rendering them hypersensitive to the cost of capital (Bistline, Mehrotra, and Wolfram 2023). Inflationary energy price shocks trigger monetary policy tightening; because renewables are far more capital-intensive than their fossil-fuel substitutes such as natural gas, the resulting rise in the cost of capital creates a differential financial barrier to clean energy adoption. The financial headwinds extend to the innovation frontier: Aghion et al. (2024) document that contractionary monetary policy reduces green patenting significantly more than non-green patenting, as tight credit disproportionately constrains the young firms at the forefront of clean innovation. This contrasts with the finding of Känzig et al. (2025) that green innovation is broadly countercyclical — driven by a “green is in the future” channel that insulates long-term R&D from short-term recessions. Second, grid-level technological constraints create interdependencies across fuels. Verdolini, Vona, and Popp (2018) document a strong complementarity between intermittent wind and solar and fast-reacting fossil-based backup capacity (primarily natural gas). Consequently, when oil shocks simultaneously disrupt natural gas supply

and spike gas prices, the adoption of solar is severely hindered. Indeed, the relationship between natural gas and renewables is complex; while the shale gas boom reduced emissions by displacing coal, Acemoglu et al. (2023) and Harstad and Holtsmark (2026) argue that cheap natural gas can create a “fossil-fuel trap” that delays clean innovation, whereas Lindequist and Selent (2025) find evidence of gas acting as a green “bridge fuel”. I contribute to this literature by documenting that oil price shocks do not accelerate clean energy adoption: solar generation declines following such events, while alternative-fuel vehicle registrations merely track overall car demand.

1.2 Organization

The remainder of this paper is organized as follows. First, section 2 studies the implications of oil price hikes on worldwide greenhouse gas concentrations. Section 3 documents the dynamics of a broad spectrum of fossil fuels. Section 4, in turn, details the mechanisms that underlie changes in emissions efficiency — particularly focusing on the extensive margin — the adoption of clean energy alternatives. Section 5 concludes.

2 Global Warming Potential of Oil Price Shocks

Oil price shocks affect greenhouse-gas emissions through various channels with potentially opposing forces. On the one hand, an increase in oil prices directly decreases demand, thereby lowering GHG emissions derived from petroleum consumption. On the other, it induces substitution dynamics, increasing the use of alternative sources, more and less polluting, such as coal and solar, respectively. The former counteracts the emissions decline from a fall in crude oil demand, the latter exacerbates it.

In this section I quantitatively evaluate the overall emissions effect of oil price shocks using standard macroeconomic time-series techniques, and thus the ensuing global warming potential changes they generate.

2.1 Data

Emissions I use monthly average global atmospheric concentration data from the National Oceanic and Atmospheric Administration (NOAA) (Lan, Tans, and Thoning 2026; Lan, Thoning, and Dlugokencky 2026) to study the global warming effects of oil price shocks. I use the four greenhouse gases (GHG) for which NOAA publishes monthly global-mean abundances from its Global Greenhouse Gas Reference Network: the three main long-term drivers of climate change — carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O) — and sulfur hexafluoride (SF_6). The remaining long-lived greenhouse gases, mostly halocarbons, are not included in the network’s data. Their omission matters little: CO_2 , N_2O and CH_4 are the three largest contributors to the 54% increase in effective radiative forcing from long-lived greenhouse gases between

1990 and 2024, with CO_2 alone accounting for about 81% of it (NOAA Global Monitoring Laboratory 2025). The data start at different dates, so I restrict my sample to the intersection of their spans. As a result, all data start in January 2001 and end in December 2025. Starting the sample in the early 21st century has the advantage of coinciding with the rise of renewables as viable energy sources, a significant development for the world economy’s energy-source choice set.

In order to measure the global warming potential of the combined gas concentrations added, I compute the carbon dioxide equivalent (CO_2e) for the 4 GHGs for which I have data. This individual index measures how much CO_2 would be necessary to increase the globe’s temperature by the same amount as that resulting from the combined GHG emissions since pre-industrial times, all else equal, i.e., keeping the other GHGs’ concentrations constant at pre-industrial levels. Radiative forcing measures how much a change in the atmospheric concentration of a gas, relative to pre-industrial times, alters the Earth’s energy balance, in watts per square metre ($W m^{-2}$): positive forcing means the planet retains more energy than it radiates back to space, warming it. Effective radiative forcing (ERF) additionally accounts for the rapid adjustments of the atmosphere — temperature, water vapour and clouds — that follow, and is the measure the IPCC recommends as the best predictor of warming.

I measure the ERF of each gas with the simplified expressions of Etminan et al. (2016), as updated by Meinshausen et al. (2020) and adopted by the IPCC (Intergovernmental Panel on Climate Change (IPCC) 2023b). Denoting the atmospheric concentrations of CO_2 , CH_4 , N_2O and SF_6 in month t by C_t , M_t , N_t and S_t , total forcing is the sum of the forcings of the individual gases,

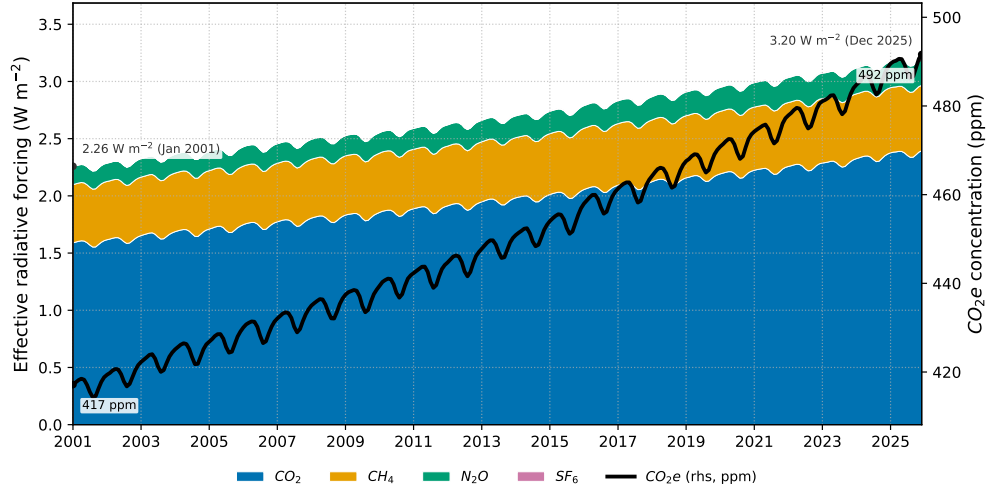
$$\Delta F(C_t, M_t, N_t, S_t) = \Delta F_{CO_2}(C_t, N_t) + \Delta F_{CH_4}(M_t, N_t) + \Delta F_{N_2O}(C_t, M_t, N_t) + \Delta F_{SF_6}(S_t), \quad (1)$$

where each term measures the forcing relative to that gas’s pre-industrial (1750) concentration — C_0 , M_0 , N_0 and S_0 (Intergovernmental Panel on Climate Change (IPCC) 2023a) — and is zero at it. The forcing of one gas can depend on the concentrations of others because their absorption bands overlap. The CO_2e concentration in month t , C_t^{eq} , is the level of CO_2 that, with all other gases held at their pre-industrial levels, produces the same total forcing as the observed concentrations. It is the unknown in

$$\Delta F(C_t^{eq}, M_0, N_0, S_0) = \Delta F(C_t, M_t, N_t, S_t), \quad (2)$$

which I invert numerically for every month. In practice, the CH_4 , N_2O and SF_6 forcings on the left-hand side are zero, since these gases do not deviate from their pre-industrial levels, and all that matters is the CO_2 forcing, $\Delta F_{CO_2}(C_t^{eq}, N_0)$. I provide the functional forms, parameters and algorithm in the supplemental appendix (section B). Figure 1 plots the effective radiative forcing of each gas over my sample, together with the resulting CO_2e concentration, the dependent

Figure 1: Effective Radiative Forcing by Gas and CO_2 -Equivalent Concentration



Notes: Stacked effective radiative forcing of CO_2 , CH_4 , N_2O and SF_6 relative to pre-industrial (1750) levels, in $W m^{-2}$ (left axis), and the resulting CO_2e concentration, in ppm (right axis, black line). Monthly global-mean NOAA data, 2001–2025. The forcing of SF_6 is too small to be visible at this scale.

variable in my main specification.

Oil Price Shocks To capture exogenous variations in the price of crude oil, I use the oil supply news shocks series provided by Känzig (2021). To generate this series, Känzig (2021) first exploits the institutional features of the Organization of Petroleum Exporting Countries (OPEC): at pre-determined dates OPEC systematically provides an outlook on their own crude oil production. Using high-frequency oil futures data, it is then possible to measure the changes in the price of crude oil futures contracts in a narrow window of time around these scheduled events. This high-frequency approach thus avoids reverse causality problems and isolates supply from demand-driven fluctuations in oil price futures. In a second step, a standard proxy vector autoregression identifies exogenous shifts in the price of crude oil due to changes in the expected supply of global crude oil by applying instrumental variables to the residuals of the oil supply equation and using the news surprise series previously generated. The resulting estimates are the exogenous shock series, used widely in the macroeconomics literature. The series is normalized so that a 1-unit shock corresponds to a 10% increase in the price of crude oil on impact.¹

Mori and Peersman (2024) document that this identification approach is subject to informational deficiencies: the identified structural innovations are found to be Granger-caused by financial variables, indicating that markets can predict OPEC decisions through information not captured by the vector auto-regressive model, thereby contaminating the instrument. They show

1. The shock series over my sample is plotted in the supplemental appendix (figure A.1).

that augmenting the econometrician’s information set with financial indicators resolves these issues and yields sharper, more stable estimates. I address this concern directly by including financial controls in my local projections, as described in subsection 2.2.

In the supplemental appendix (section A), I compare the oil supply news shocks with the structural oil supply shocks of Baumeister and Hamilton (2019), and explain why I use the former.

2.2 Methodology

To assess the effect of oil price shocks, I resort to the local projections methodology popularized by Jordà (2005). For $h = 0, \dots, H$, I estimate the long-differences model:

$$\Delta^{h+1}y_{t+h} = \alpha_h + \alpha_{m(t),h} + \beta_h \cdot s_t + \sum_{k=1}^{12} (\rho_{h,k} y_{t-k} + \Gamma'_{h,k} X_{t-k} + \tau_{h,k} s_{t-k}) \quad (3)$$

where α_h is an intercept, to account for the average growth at horizon h . $\alpha_{m(t),h}$ are month fixed-effects aimed at capturing the seasonal patterns present in both greenhouse gas emissions and oil prices. $m(t)$ is the calendar month at date t . s_t is the exogenous oil price shock and X_{t-k} are control variables. $\Delta^{h+1}y_{t+h} = y_{t+h} - y_{t-1}$ is the difference between the dependent variable y at horizon h relative to the month before the shock $t - 1$, and is either the concentration of individual GHG in the atmosphere, or of my measure of equivalent CO_2 . Following Montiel Olea and Plagborg-Møller (2021), I include 12 lags of the dependent variable and compute confidence bands using a wild bootstrap based on Liu (1988). The former show that combining lags of the dependent variable with heteroskedasticity-robust standard errors is a sufficient approach for local projection inference. I rely on Mammen’s (Mammen 1993) distribution to sample the bootstrap multipliers².

As discussed in the previous subsection, Mori and Peersman (2024) show that financial variables Granger-cause the oil supply news shocks, contaminating the instrument through anticipation effects. I address this by including 12 lags of the shock itself and of 2 financial controls: the global financial cycle of Miranda-Agrippino and Rey (2020), which captures broad movements in global risk appetite and cross-border financial conditions, and the financial uncertainty index of Jurado, Ludvigson, and Ng (2015), which measures the unpredictability of financial variables, reflecting forward-looking information that market participants embed in asset prices ahead of OPEC decisions. I keep this parsimonious specification uniform across all empirical sections of the paper, including the panel analyses where the available time span is considerably shorter, so that results remain comparable throughout.

The sample includes the COVID pandemic period. To handle this, I follow Baumeister and Hamilton (2025) and drop the year of 2020 completely from the data. Because of the inclusion of

2. Details on the algorithm and the multiplier distribution are provided in the supplemental appendix (section C).

lags, 2021 is also effectively dropped.

2.3 Results

2.3.1 The Macroeconomic Backdrop

Before turning to the emissions results, figure 2 documents the response of key macroeconomic aggregates to the shock. This serves two purposes: it validates the shock identification for the restricted sample, 2001 to 2024 — excluding 2020, and it establishes the macroeconomic context within which all results should be interpreted. This exercise also helps to clarify my goal: I am not analyzing a pure decline in oil prices while everything else is kept constant — a useful theoretical question but one with less practical importance when the events that generate such oil price fluctuations also affect macroeconomic dynamics — even though they originate from the crude oil market alone. I am precisely interested in the joint dynamics induced by these events, coined “oil shocks”.

Panel *a*) of figure 2 shows the real oil price, computed by deflating the West Texas Intermediate price by the US consumer price index, both obtained from the Federal Reserve Economic Data (FRED) database. It rises by about 10% on impact — in line with the normalisation of Känzig (2021) — and gradually decays, becoming statistically indistinguishable from zero at around 8 months, with point estimates centering at zero 1 year after — much less persistent than what Känzig (2021) finds.

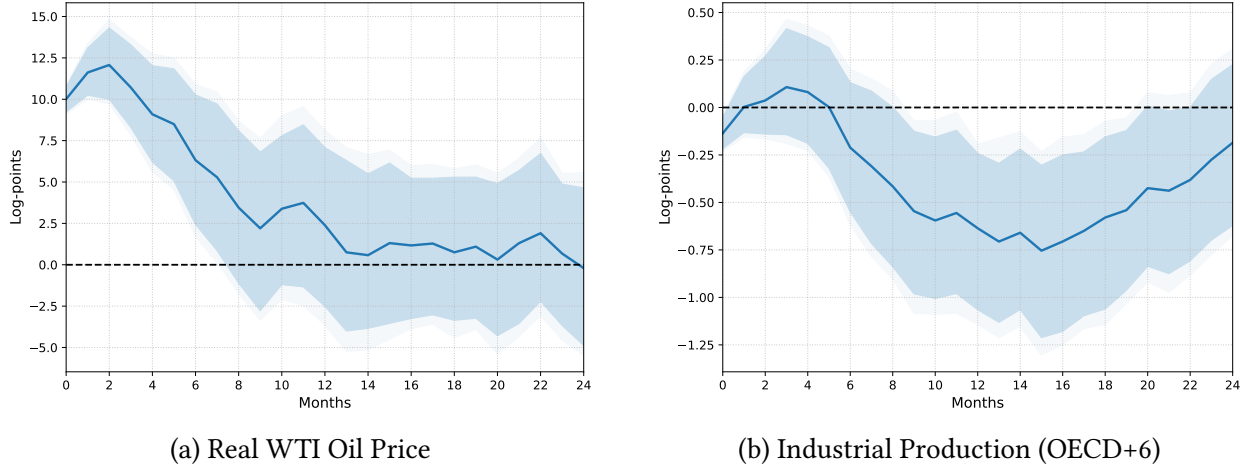
Panel *b*) of figure 2 confirms that the shock is contractionary: industrial production for OECD and 6 selected countries³, obtained from Baumeister and Hamilton (2019), is little changed in the first months, then declines and remains below trend at the 2-year horizon, consistent with a negative aggregate supply shock. The drop peaks after 15 months, implying a 0.75% drop in global economic activity. The peak is almost the same as in Känzig (2021). Distinctly, it is much less persistent, in line with the dynamics of the WTI.

2.3.2 Global Warming Potential

The backdrop surrounding oil news shocks has two effects on energy consumption: first, the slowdown in economic activity decreases overall demand for energy. Second, the increase in crude oil prices incentivizes substitution towards alternative energy sources. On the one hand, this effect can be negative, inducing more coal use — the most polluting fossil energy source. On the other, it can accelerate the uptake of renewable energy sources, such as solar. Naturally the horizons of these two substitution effects differ. The first could play out from the onset of the

3. Industrial production is used as a proxy for economic activity because it is available at the monthly frequency, unlike gross domestic product.

Figure 2: Macroeconomic Responses to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable in each panel is the $h + 1$ long-difference in the respective series. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

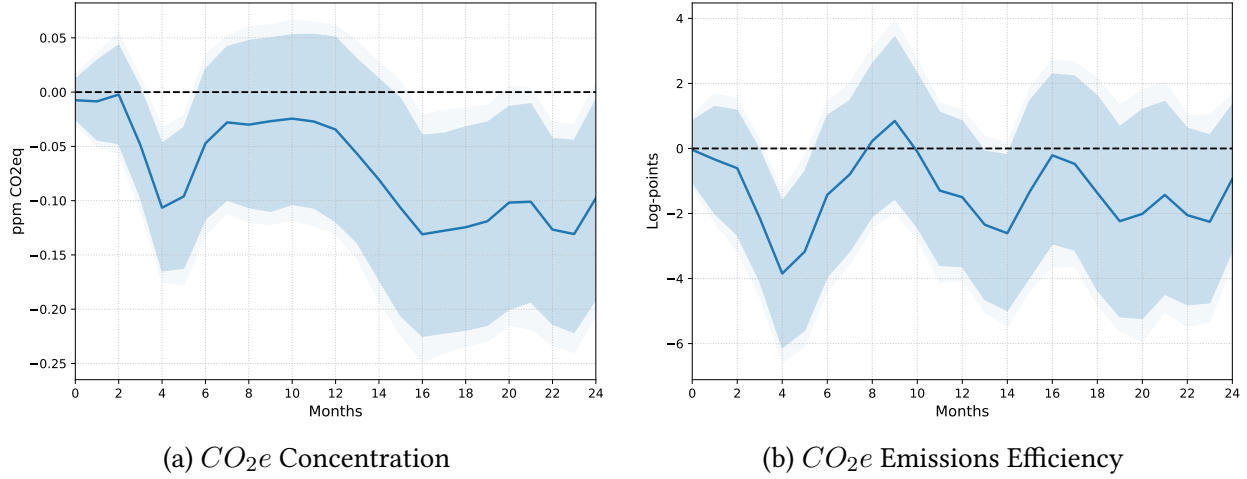
shock as coal plants with available capacity increase their output. The second can potentially take longer as it requires investments in new infrastructure since renewables tend to work at the limit of their useful capacity.

As a result, the net quantitative effect of oil price shocks on overall GHG emissions is not ex-ante determined. To assess it, panel *a*) of figure 3 presents the impulse response functions for the aggregate CO_2e index, estimated using equation (3). Following an oil supply news shock, atmospheric CO_2e declines by 0.11 parts per million (ppm) relative to trend after 4 months. Following a short rebound, the concentration decline persists and stabilizes at 0.10 ppm. This is statistically significant evidence that oil price increases lower greenhouse-gas concentrations, and thus global warming potential, relative to trend. This decline is only marginal when compared to long-run trends: it is equivalent to 3.24% of the average yearly increase in atmospheric CO_2e concentration, when measured over my sample, from 2001 to 2025.

While there is a decline in emissions, it may be a consequence of a decline in economic activity and not of increased dirty energy efficiency, i.e., in the amount of dirty energy necessary to support a certain amount of economic activity. To evaluate this, I compute the impulse response function for the logarithm of the ratio of the year-on-year difference in atmospheric CO_2e concentration over the OECD+6 industrial production index of Baumeister and Hamilton (2019),

$$\log \left(\left(\Delta^{12} CO_2e_t \right) / IP_t \right).$$

Figure 3: Response of CO_2 -Equivalent Concentration to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panel (a): $h+1$ long-difference in atmospheric CO_2e concentration (ppm). Panel (b): $h+1$ long-difference in the log ratio of the 12-month difference in CO_2e concentration to the OECD+6 industrial production index. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024 in panel (a), and January 2003 – December 2024 in panel (b); outcome data extends to December 2025.

I use year-on-year differences instead of the month-on-month flow to deal with the seasonality in atmospheric GHG concentration.

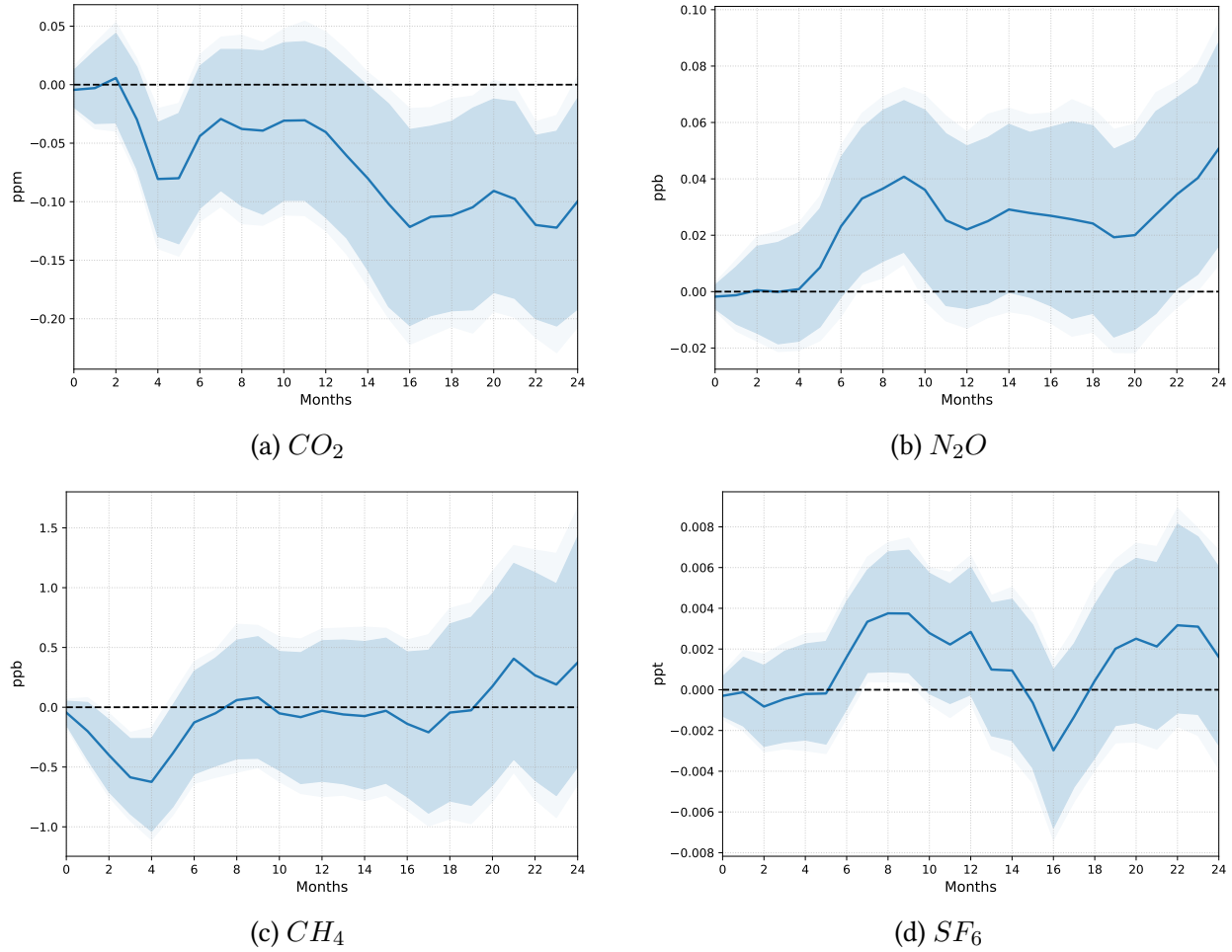
The results are plotted in panel b) of figure 3 and show a fast but short-lived decline in emissions per unit of economic activity, mostly focused on months 3 to 6. Emissions efficiency fluctuates thereafter and drops again on the 14th month. The dynamics are similar if I instead divide by industrial production summed over the same 12 months, so that both numerator and denominator are annual flows, as shown in the supplemental appendix (section D).

Overall, these results indicate that the observed decline in emissions is not just a result of the ensuing economic slowdown but of increased emissions efficiency, in line with what Känzig and Williamson (2024) find for the United States. In contrast, I cannot statistically ascertain that the efficiency effect is permanent or even long-lived beyond the time oil prices mean-revert. This discrepancy may reflect sample selection. The shocks from Känzig (2021), when used in a longer sample like in Känzig and Williamson (2024), are known to produce a more persistent decline in crude oil prices, which mechanically should sustain the efficiency incentive over a longer horizon.

2.3.3 Individual Emissions

While overall global warming potential decreases as a result of an oil price shock, this behavior may not be homogeneous across different gases because of the ensuing energy mix reshuffling.

Figure 4: Response of Individual Greenhouse Gas Concentrations to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in atmospheric concentration. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Moreover, understanding the individual GHG dynamics is important because they have different atmospheric lifetimes, and so different global warming potentials depending on the horizon of analysis (Pachauri, Mayer, and Intergovernmental Panel on Climate Change 2015).

To verify this, I also study the dynamics of each individual GHG, plotted in figure 4. The results confirm the heterogeneous dynamics. The most abundant GHG in the atmosphere, CO_2 , follows a similar path to CO_2e : first it declines to 0.08 ppm and then plateaus at 0.10 ppm. Similarly, CH_4 decreases by 0.62 parts per billion (ppb) after 4 months but rebounds and its concentration mean-reverts, becoming statistically insignificant after 8 months.

Working in the opposite direction, the most powerful GHGs rise: SF_6 increases 6 months

after the shock, but then reverts to the mean after 1 year. In the same tone, N_2O increases after 4 months, reaching 0.04 ppb by month 8, and remains elevated, at about 0.05 ppb after 2 years. The persistent elevation of N_2O is consistent with the increase in bioenergy generation documented in section 3, and with the shift in the coal mix towards lignite on impact: both biomass and lignite emit more N_2O per unit of energy than the fuels they displace.

The documented heterogeneous dynamics across GHGs motivate my study of hydrocarbons in section 3.

2.4 Robustness

I estimate several alternative specifications to ensure that my main results are robust. In particular, I deviate from the standard lag-length and consider including fewer (more) lags — for the lagged variables included in the right-hand side of equation (3) — to account for potentially shorter (longer) auto-correlation structures. I also exclude the oil news shock lags to show that the results are not purely driven by noise. Finally, to ensure that no outlier shocks drive the dynamics, I trim the shock to the 5th and 95th percentiles of the sample distribution, from 2001 to 2025, excluding any shocks outside this range from the estimation⁴.

I compare the alternative specifications' point estimates with my baseline impulse responses for CO_2e and for the emissions-efficiency measure of figure 3 in figure 5 and confirm that the point estimates are very similar across different specifications. The same comparisons are done for the other series with similar conclusions. Their plots are provided in figure D.1.

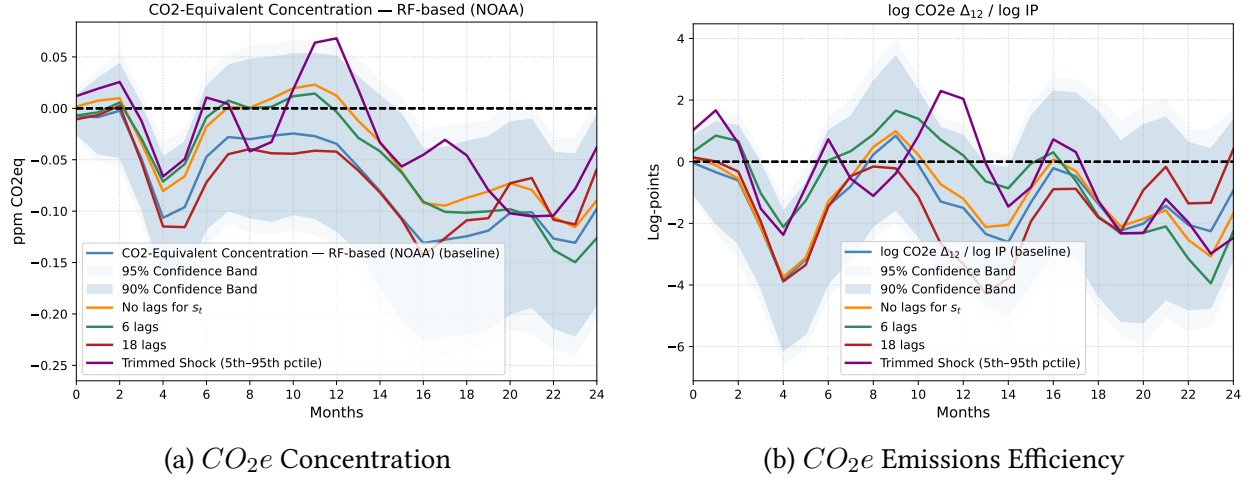
I also test for asymmetric dynamics, i.e., whether the impulse responses differ between positive (oil price-increasing) and negative (oil price-decreasing) shocks, by adding a new term $\gamma s_t \times 1_{\{s_t > 0\}}$. Figure D.2 in the supplemental appendix plots the resulting impulse responses for γ . The differential response of CO_2e does not statistically differ from zero. In contrast, after 1.5 years, emissions per unit of industrial production fall by less following a positive shock than they rise following a negative one.

3 Hydrocarbon Dynamics

In this section I study the sources of GHG that explain the concentration dynamics presented in section 2. I focus on primary dirty energy consumption. This largely includes industrial processes and energy production, which together represented about 84% of all GHG emissions worldwide (Climate Watch 2021). I thus exclude other sources of emissions like land use and agriculture. Specifically, I inspect the consumption and/or production dynamics of all types of fossil fuels following oil price hikes — less documented in the literature studying oil shocks. Over-

4. I keep the shock's lags unrestricted.

Figure 5: Robustness of the CO_2e and Emissions-Efficiency Responses to Alternative Specifications



Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from four alternative specifications: no shock lags, 6 and 18 lags of all lagged variables (dependent variable, controls and shock), and the shock trimmed to its 5th–95th percentiles. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024 in panel (a), and January 2003 – December 2024 in panel (b); outcome data extends to December 2025.

all, petroleum and natural gas consumption, and mainstream coal production decline, while the production of lignite — the dirtiest, most carbon-intensive coal — temporarily rises.

3.1 Data

I build an unbalanced monthly panel dataset of countries from a variety of sources. All data are restricted to 2001 onwards or later, depending on its availability⁵.

First, I obtain fuel consumption data from the Joint Organisations Data Initiative (JODI), a consortium that monitors and aggregates oil and gas data from national and international energy organizations. I obtain information on national refined oil fuel consumption for gasoline, diesel, jet fuel, residual oil — including bunker oil, mainly used for maritime shipping — and liquefied petroleum gas (LPG). These series thus cover primary energy derived from oil and gas (excluding electricity) used in the residential sector — specifically for heating and cooking — as well as the transportation sector, including automotive, aviation, and maritime transport. The data run from January 2002 to December 2025 and cover 106 countries. JODI demand for oil products is a derived quantity: it is not directly measured but computed as a residual from the supply-side balance⁶. Errors in any balance component can therefore occasionally produce negative demand

5. Additional details about each dataset, its coverage and the raw time-series are provided in the supplemental appendix (section E).

6. Specifically: Demand = Refinery Output + Receipts + Imports – Exports – Products Transferred +

values, which I remove from the dataset. These observations would be excluded regardless, since all dependent variables enter in log-points.

Secondly, I obtain natural gas consumption data from the same data consortium, both for the whole economy and the electricity and heat generation sectors in particular. This dataset starts later, in January 2009, and includes 80 countries. As with oil products, I remove the occasional negative values reported.

Thirdly, in the absence of high-frequency consumption data, I obtain quarterly coal production data from the International Energy Agency. It has a much shorter time-span, starting in January 2016, but covers 33 countries, a great share of the world’s production including China, Russia, and OECD countries. As with crude oil, coal consumption and production do not match exactly, since producers and consumers adjust precautionary stockpiles, but production remains a good proxy for consumption.

3.2 Methodology

Ideally, I would build aggregate measures for each of the variables from the data collected. Nevertheless, the staggered inclusion of countries prevents such an approach. A uniform subsample would either not be entirely representative of global activity or would have a much shorter time-span. Additionally, a panel set-up can increase the statistical power of my estimates in place of shorter series as is the case for coal, natural gas or for the power sector data. As a result, I use the individual country observations in the estimation. Following Chodorow-Reich, Nenov, and Simsek (2021) and Ottonello and Winberry (2020), I adapt the methodology from subsection 2.2 to the panel set-up by running a fixed-effect version of local projections:

$$\Delta^{h+1} y_{i,t+h} = \alpha_{i,m(t),h} + \beta_h s_t + \sum_{k=1}^{12} (\rho_{h,k} y_{i,t-k} + \Gamma'_{h,k} X_{t-k} + \kappa_{h,k} \Delta C_{i,t-k} + \tau_{h,k} s_{t-k}) \quad (4)$$

where i indexes the country. $\alpha_{i,m(t),h}$ are country-specific calendar-month fixed effects. They nest country fixed effects, and so account for differential growth trends across countries, while also capturing the inherent seasonality of the dependent variable, which may vary by country, especially because of different climates and geographies. X_{t-k} are the same global controls included in equation (3), and $\Delta C_{i,t-k}$ are lags of the logarithmic growth rate of country i ’s nominal exchange rate in USD from the International Monetary Fund. This country-specific control partially accounts for time variation in countries’ economic and financial conditions and has a very broad coverage.⁷ All the dependent variables are defined in log-points.

All regressions are estimated via weighted least squares, so that β_h can be interpreted as a

Interproduct Transfers – Stock Change – Statistical Difference.

7. Only one country, Cuba, lacks exchange-rate data entirely, and drops out of the refined oil sample.

representative global estimate (Ramey 2019). I use the lagged raw dependent variable as weight, unless otherwise specified. Compared to static weights, using time-varying weights accounts for differential trends across countries’ energy use patterns. Using a yearly moving average instead, to decrease the influence of seasonality, does not materially change the results, as shown in the supplemental appendix (subsection F.5). Neither does dropping the lags of the shock, replacing the country-month fixed effects with additive country and month fixed effects, or trimming the most extreme shock realisations. The appendix also plots the pre-trends of every panel regression, which show no significant dynamics ahead of the shock, so the responses do not reflect pre-existing trends that the specification fails to capture.

Following Almuzara and Sancibrián (2026), I cluster standard errors by time period: every country is exposed to the same aggregate oil shock at each date, which creates cross-sectional dependence across countries within a period. As in the time-series regressions, the lags of the dependent variable are included in equation (4) to capture the potential autocorrelation.

3.3 Refined Oil

I first plot the impulse response functions for refined oil products consumption in figure 6. All regressions are weighted by the lag of the respective raw petroleum variable. Consumption drops overall, albeit at different moments in time. Well documented in the literature, retail fuels, diesel and gasoline, drop almost instantaneously. In contrast, and less studied, liquefied petroleum gas, residual fuel, and jet kerosene decline only later. LPG declines first, after about half a year, while residual fuel and jet kerosene decline only more than a year after the shock hits.

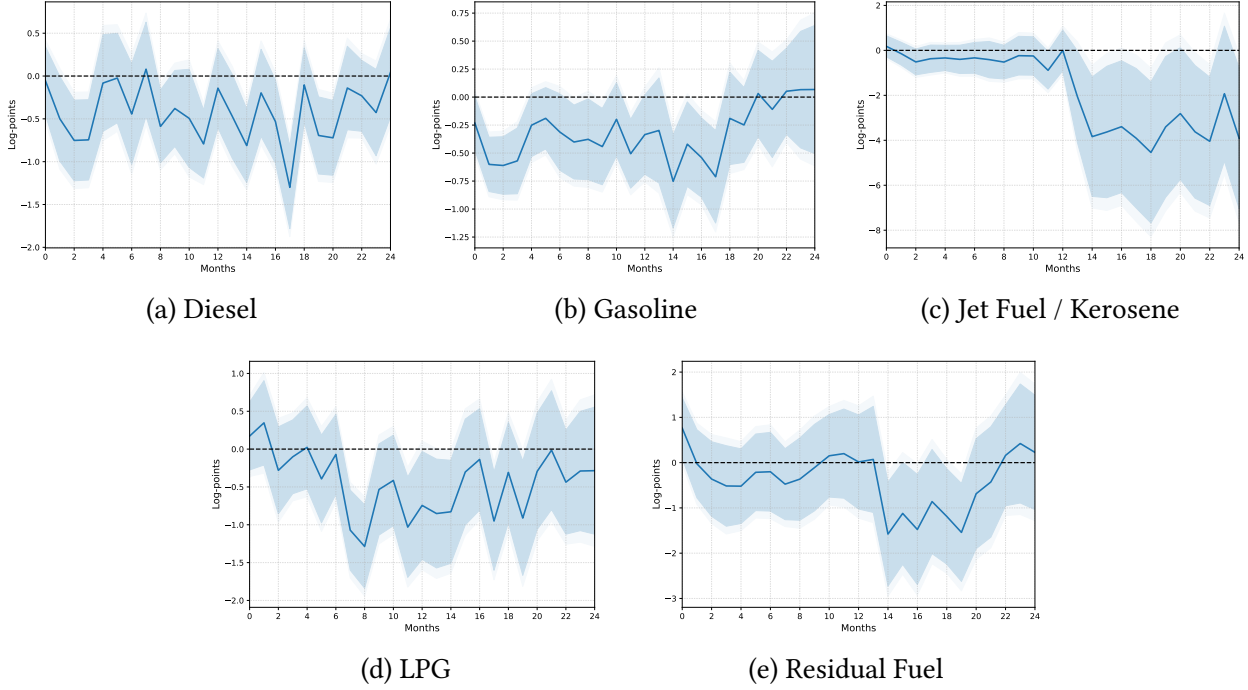
Their semi-elasticities to crude oil prices differ too. Diesel and gasoline decline almost immediately, by about 0.61 percentage points after 2 months following an oil shock that raises crude oil prices by 10% on impact, but revert to trend within 2 years. LPG declines by 1.07 percentage points by month 7, while residual fuel decreases by 1.58 percentage points by month 14. Both revert to the trend within 2 years. Differently, jet fuel sees the biggest drop of 3.84 percentage points, only after 14 months, and does not recover within the horizon considered.

3.4 Natural Gas

Natural gas is a partial substitute for oil in power generation and industrial heating, so an oil price increase may divert demand toward gas (Baumeister et al. 2024). At the same time, oil and gas are typically extracted from the same wells (U.S. Energy Information Administration 2013). As a result, disruptions in the supply of the former are known to have consequences on the supply of the latter. Moreover, reduced economic activity decreases the demand for natural gas.

To determine which channel dominates, I compare the dynamics of consumption and prices.

Figure 6: Response of Refined Oil Consumption to an Oil Supply News Shock



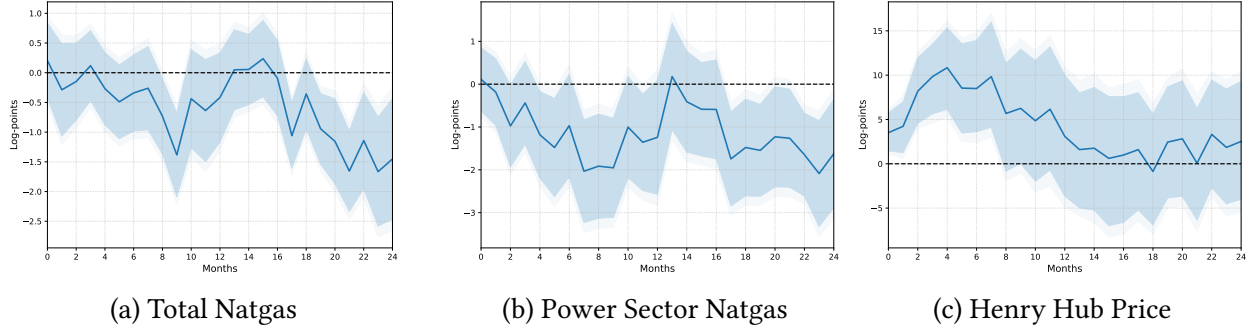
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the log of monthly national consumption (thousand barrels). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective consumption series (thousand barrels). COVID-19 (2020) is excluded from the sample. Shocks: January 2003 – December 2024; outcome data extends to December 2025.

Figure 7 presents the impulse response functions for natural gas consumption following an oil supply news shock. Natural gas declines in general, and in the power sector specifically. Total consumption declines, briefly returns to trend around month 14, and stands 1.45 percentage points below trend at the 2-year horizon. The drop is more severe in the power sector, which declines by 1.62 percentage points relative to trend at the 2-year horizon, with only a partial rebound around month 14. This drop in natural gas consumption is accompanied by an increase in gas prices. The Henry Hub reference gas price, obtained from FRED, plotted in panel *c*) of figure 7 rises by 3.51 percentage points on impact, reaching a peak of 11 percentage points after 4 months before slowly declining. This suggests that adverse supply effects dominate both the demand-side substitution and income effects, explaining the dynamics I observe in consumption.

3.5 Coal

Coal is the biggest source of pollution in primary energy production. It is a critical input in industrial production and a strong substitute for natural gas in power generation. To understand

Figure 7: Response of Natural Gas Consumption to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panels (a)–(b): the dependent variable is the $h + 1$ long-difference in log monthly national natural gas consumption (m^3). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective natural gas consumption series (m^3). COVID-19 (2020) is excluded from the sample. Shocks: January 2010 – December 2024; outcome data extends to December 2025. Panel (c): time-series local projection, equation (3); the dependent variable is the $h + 1$ long-difference in the log real Henry Hub natural gas spot price (FRED, deflated by the US consumer price index), with 90% and 95% confidence bands based on heteroskedasticity-robust bootstrapped standard errors. Shocks: January 2002 – December 2024.

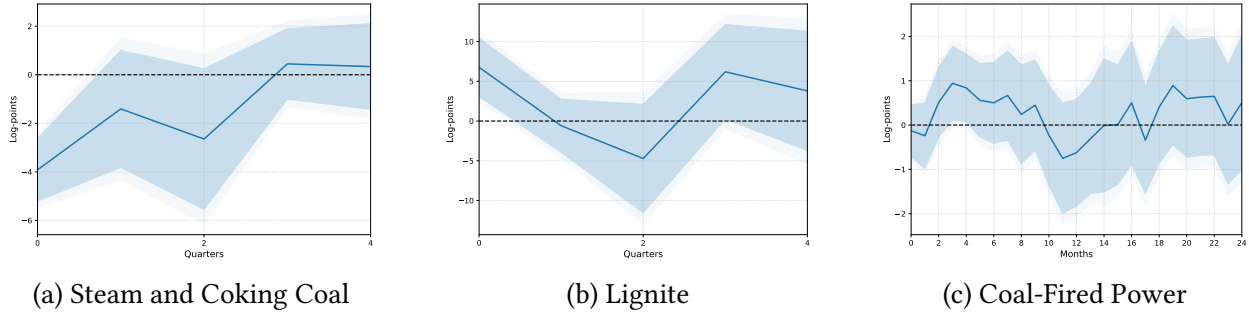
its dynamics following an oil price shock, I adapt the methodology presented in subsection 3.2 to the data’s quarterly frequency. Because of the different uses, I separate the main results by type of coal: steam and coking, and lignite, and leave the results for total coal production in the supplemental appendix (subsection F.1). Although steam coal is used primarily for electricity and heating, and coking coal is primarily used in an industrial setting — mainly metallurgy — the data report their production jointly, so I cannot assess their dynamics separately.

Panels *a*) and *b*) of figure 8 present the impulse response functions for each type of coal. I consider only a 4-quarter horizon since the sample is much shorter, for most countries starting only in 2016. Steam and coking coal decrease by 3.92 percentage points on impact, but recover after 3 quarters. In contrast, lignite, the lowest-ranked and most carbon-intensive coal, increases by 6.76 percentage points on impact, but the increase reverses within two quarters. Overall, total coal production follows the dynamics of steam and coking coal, and decreases by 3.18 percentage points on impact, staying depressed for 6 months before recovering.

Because I am unable to separate coking from steam coal, I cannot draw conclusions regarding coal-based electricity production from these figures. To ascertain its net response, panel *c*) of figure 8 plots instead the monthly impulse response functions for coal-based electricity production⁸. It shows evidence of a temporary rise in the first 3 months, before mean-reverting, in line with the lignite dynamics. This indicates that the latter’s increase compensates for the potential fall in steam coal, and helps explain why N_2O emissions rise, as shown in subsection 2.3. Lignite can

8. Details about the data are presented in subsection 4.2.

Figure 8: Response of Coal Production and Coal-Fired Electricity Generation to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panels (a)–(b): the dependent variable is the $h + 1$ long-difference in log quarterly coal production; the horizon is measured in quarters. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-quarter. Regressions are weighted by the lag of the raw production level. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to July 2025. Panel (c): the dependent variable is the $h + 1$ long-difference in log monthly coal-fired electricity generation (TWh). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of coal-fired electricity generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

produce 18% to 82% more nitrous oxide than bituminous coal⁹ (Government of Canada 2024). As such, while I do not observe a sustained increase in coal-based electricity production, the change in the underlying coal mix contributes to the rise in N_2O -emissions intensity.

3.6 Bioenergy

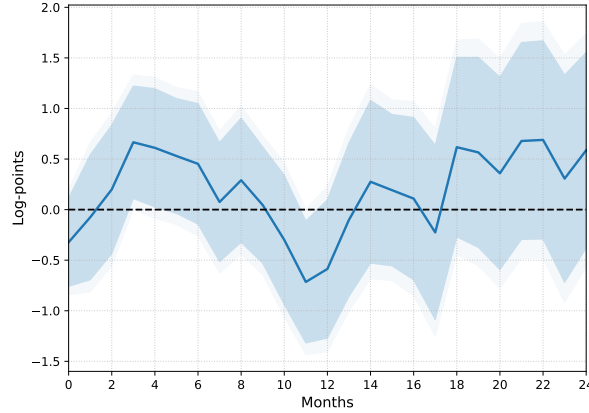
Bioenergy — electricity generated from burning organic material such as wood pellets, agricultural residues, and dedicated energy crops — occupies an intermediate position between fossil fuels and clean renewables. Unlike solar and wind, it is dispatchable and emits GHGs during combustion, including CO_2 and N_2O . Per unit of energy, burning solid biomass emits far more N_2O than burning natural gas (Eggleston et al. 2006), the dispatchable fuel whose consumption falls following the shock¹⁰.

Figure 9 presents the impulse response of bioenergy generation. Following the shock, bioenergy rises, reaching 0.59 percentage points at the 2-year horizon. Though the bands are wider, certain horizons are statistically significant at conventional levels, confirming the elevated trend in biomass usage. This is consistent with the substitution dynamics in the dispatchable segment of

9. Government of Canada (2024) estimate that lignite produces 2 g of N_2O per GJ of energy generated when used for utility level electricity whereas sub-bituminous and bituminous produce 1.1 to 1.7 g/GJ.

10. The IPCC default emission factors for stationary combustion in the energy industries are 4 kg of N_2O per TJ for solid biomass (wood and wood waste, and other primary solid biomass), against 0.1 kg/TJ for natural gas (Eggleston et al. 2006, Volume 2, Chapter 2, Table 2.2).

Figure 9: Response of Bioenergy Generation to an Oil Supply News Shock (TWh)



(a) Bioenergy Generation (TWh)

Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly bioenergy generation (TWh). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of bioenergy generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

power generation, as the prices of coal and natural gas rise. This increase in biomass combustion further contributes to the persistent elevation of atmospheric N_2O concentration documented in section 2.

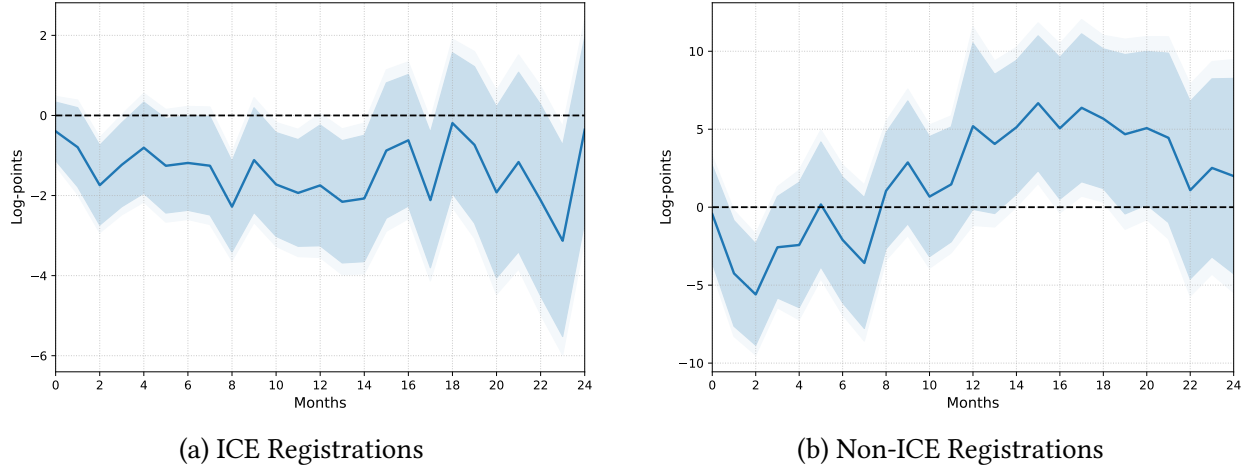
4 Jump-Starting The Energy Transition

The increase in oil and gas prices is typically viewed as an opportunity for electrification and for the adoption of alternative energy sources, in particular, renewable energy sources. In this section, I study those dynamics. Specifically, I focus on the behavior of (private) road transportation, the biggest petroleum-consuming sector (International Energy Agency 2023), and electricity generation. Currently, these are the two major energy-consumption activities with the widest and most effective set of alternative technologies – to fossil fuels – available. Notwithstanding, I find no evidence of an acceleration towards cleaner energy alternatives as a result of oil shocks – in line with the purely temporary increase in emissions-efficiency documented in subsubsection 2.3.2.

4.1 Road Transportation

I focus on the private vehicle market. On top of data availability, personal lower-emission vehicles are much more developed than alternative transportation sectors, such as aviation or shipping, and so can be considered viable alternatives to internal combustion engine (ICE) cars.

Figure 10: Response of Vehicle Registrations to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective vehicle registration count. Non-ICE includes zero-emission vehicles (battery electric, hydrogen), plug-in and non-plug-in hybrids, and alternative-fuel ICE vehicles (LPG, ethanol-only); flex-fuel cars are counted as ICE. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025. Total registrations (ICE and non-ICE combined) are reported separately in the supplemental appendix (figure F.5).

To study how non-ICE vehicles’ attractiveness changes following increases in oil prices, I use vehicle registration data from Andrew (2026), who collects monthly new registrations from national statistical registries. Where a country also reports registrations including used vehicles, I keep only new registrations, so that the data reflect new additions to the fleet and exclude used-vehicle imports. The earliest records start in January 1960, but coverage is sparse until the 2010s: most of the 52 countries in the sample enter between 2010 and 2021. I repeat the monthly panel methodology described in subsection 3.2 and start the sample in 2001 to be in line with the remaining exercises.

Figure 10 plots the impulse responses of standard ICE registrations alongside all non-ICE registrations — a composite category including zero-emission vehicles (ZEV) — battery electric and hydrogen — plug-in and non-plug-in hybrids, and alternative-fuel ICE vehicles running on LPG or ethanol only.¹¹ ICE registrations fall clearly and persistently, reaching a maximum decline of 2.28 percentage points 8 months after the shock and remaining depressed over the 2-year horizon, consistent with consumers delaying or cancelling purchases of high-operating-cost vehicles.

Non-ICE registrations display similar dynamics at first, with registrations falling by 5.60 per-

11. Flex-fuel (ethanol/petrol) cars, which make up most of Brazil’s registrations, can run entirely on petrol and are therefore counted as ICE. Mild hybrids are also counted as ICE.

centage points after 2 months. As I show in the supplemental appendix (subsection F.4), this decline happens across zero-emission and hybrid vehicles, but barely affects alternative-fuel ICE vehicles, such as LPG and ethanol-only cars. As a result, the percentage of new non-ICE registrations actually falls at first.

After an initial fall, non-ICE registrations recover and rise above trend, reaching a peak of 6.67 percentage points after 15 months. In contrast, this overshoot is driven mostly by ZEVs and hybrids, while alternative-fuel ICE cars increase faster, starting 6 months after the shock, but also mean-revert earlier — while the former two categories continue to increase. This rebound is actually not sufficiently large for me to observe a statistically significant increase in the percentage of non-ICE registrations — so this means that the rebound just follows the dynamics of overall vehicle purchases; it does not represent an accelerated transition towards alternative energy sources in private road vehicles.

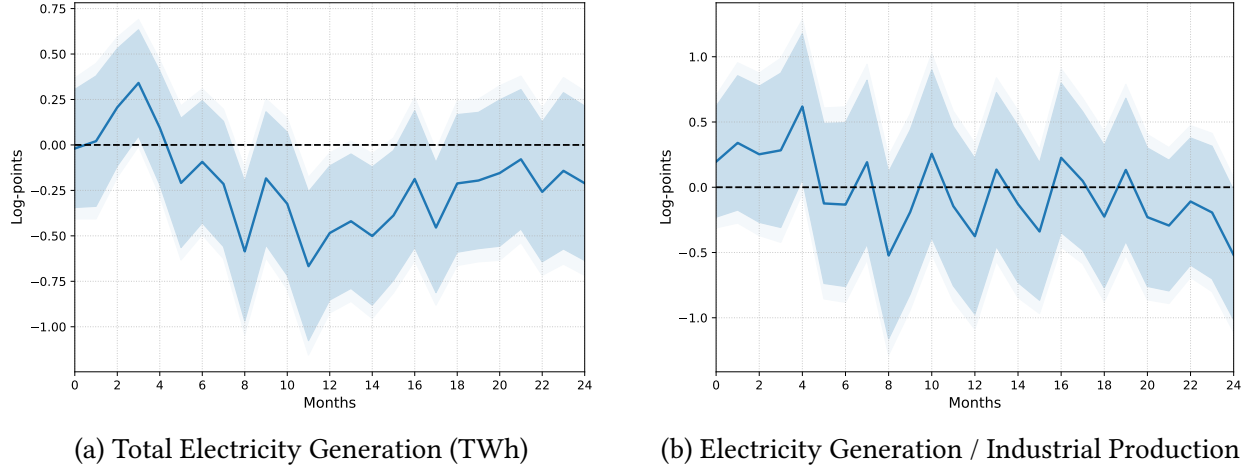
4.2 The Power Sector

Two opposing forces shape the electricity sector’s response to an oil supply shock. First, the contraction in economic activity documented above suppresses aggregate energy demand, pulling total electricity generation down. Second, higher oil prices compress the competitive advantage of oil-intensive processes and raise the relative attractiveness of electrification and renewable generation in particular. These substitution incentives operate at two margins: the intensive margin, where existing power plants run closer to installed capacity, and the extensive margin, where higher expected returns accelerate investment in new infrastructure. Section 3 highlighted the former, whereby lignite coal production rises and natural gas consumption decreases in the power sector. This section focuses on the latter.

Data on electricity generation by source, and on wind and solar installed capacity, come from Ember, a think tank that compiles data from national and international energy organizations. I obtain monthly TWh of electricity generated from coal, gas and other fossil fuels, nuclear, bioenergy, wind, solar, hydro, and other renewable sources, covering 85 countries.¹² The data start in December 1998, but the bulk of countries enter between 2015 and 2019. Capacity data record GW of nameplate capacity for wind and solar installations — including privately owned rooftop solar — starting only in January 2016 and covering 25 countries. Because these data do not account for depreciation and have such short and narrow coverage, I only use them as a robustness check, and instead focus on electricity generation. In the supplemental appendix (subsection E.1), I confirm that these data match well the annual reporting from the US Energy Information Administration (EIA).

12. Vietnam is excluded from both the generation and capacity samples, as Ember itself flags its reporting as intermittent and unreliable.

Figure 11: Response of Total Electricity Generation to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panel (a): the dependent variable is the $h + 1$ long-difference in log monthly total electricity generation (TWh). Panel (b): the dependent variable is the $h + 1$ long-difference in the log ratio of monthly electricity generation to the industrial production index. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Panel (a) is weighted by the lag of total electricity generation (TWh); panel (b) is weighted by the previous year's real GDP. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

4.2.1 Total Electricity Generation

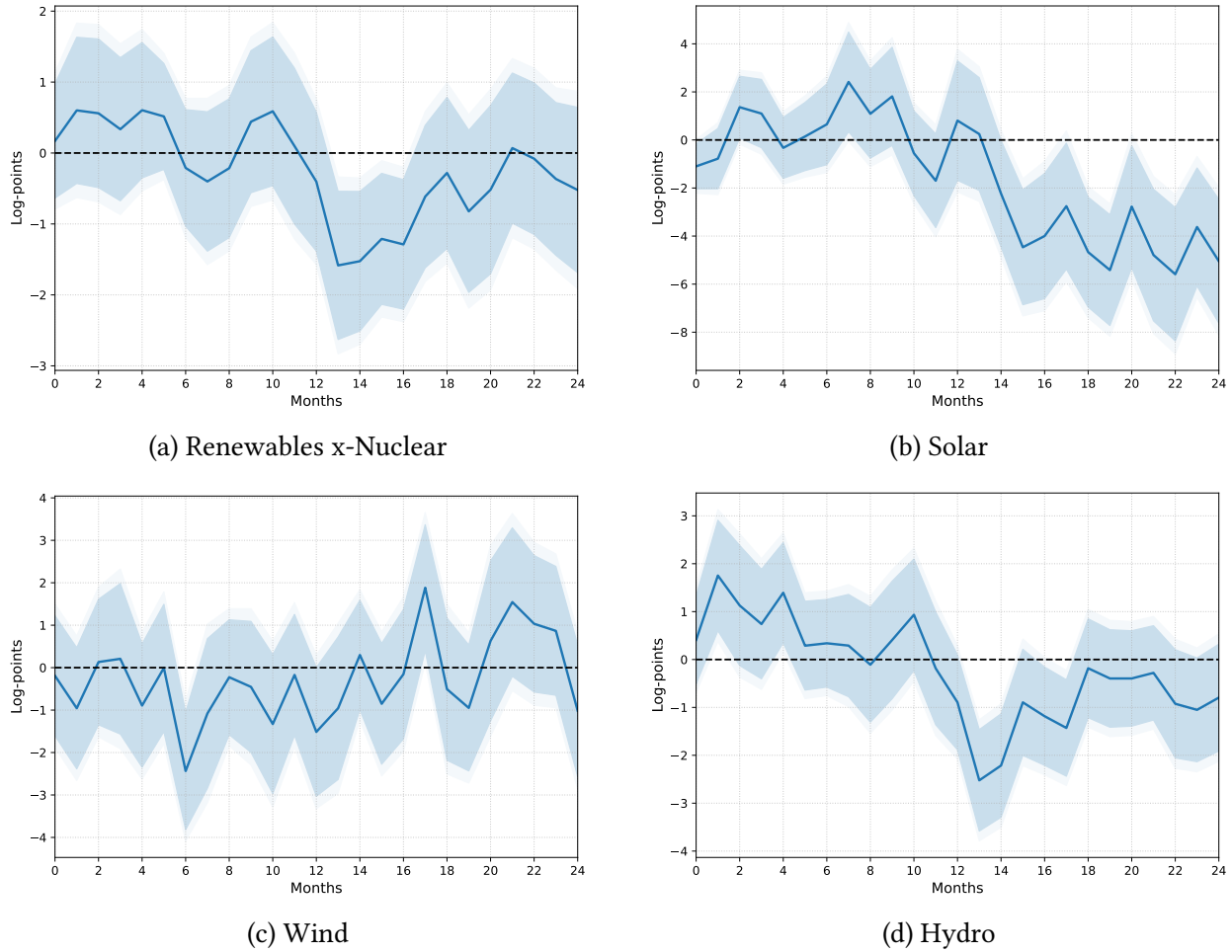
I start by inspecting how electricity consumption changes following oil price shocks. Figure 11 presents the impulse response functions for total electricity generation and for the ratio of electricity generation to industrial production. The latter is weighted by the previous year's real GDP to ensure comparability across countries – since industrial production is a country-specific index.¹³ Total generation contracts, with the drop peaking at 0.59 percentage points. The electricity-to-industrial production ratio confirms that electricity efficiency – the inverse of my ratio, output per TWh – does not increase. In particular, it does not change significantly throughout. As a result, I find no evidence that electricity steps in for other sources of secondary energy.

4.2.2 Clean energy sources

Figure 12 plots the impulse response functions for clean electricity generation. Total renewable generation – wind, solar, hydro and other renewables, excluding bioenergy – decreases roughly a year after the shock by 1.58 percentage points and stabilizes at 0.52 percentage points at a 2-year

13. Because each country's industrial production index has its own base year, the level of this ratio is not comparable across countries. This does not affect the estimates: the base year enters the ratio as a country-specific constant, which cancels in the long-difference dependent variable and, in the lagged levels, is absorbed by the country fixed effects.

Figure 12: Response of Renewable Electricity Generation by Source to an Oil Supply News Shock (TWh)



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly electricity generation (TWh) from the respective source. Renewables x-Nuclear groups wind, solar, hydro and other renewables, and excludes bioenergy. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective generation series (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

horizon.

This result appears driven almost entirely by the decline in solar generation, which collapses after 1 year of steady production, standing 5.05 percentage points below its trend after 2 years. I show in the supplemental appendix (subsection F.2) that this decline surpasses the total electricity fall, significantly so towards the full 2-year horizon. In the supplemental appendix (subsection F.3), I observe very similar dynamics for solar nameplate capacity, with the difference that the decline relative to trend occurs much faster — potentially due to the much shorter and narrower panel.

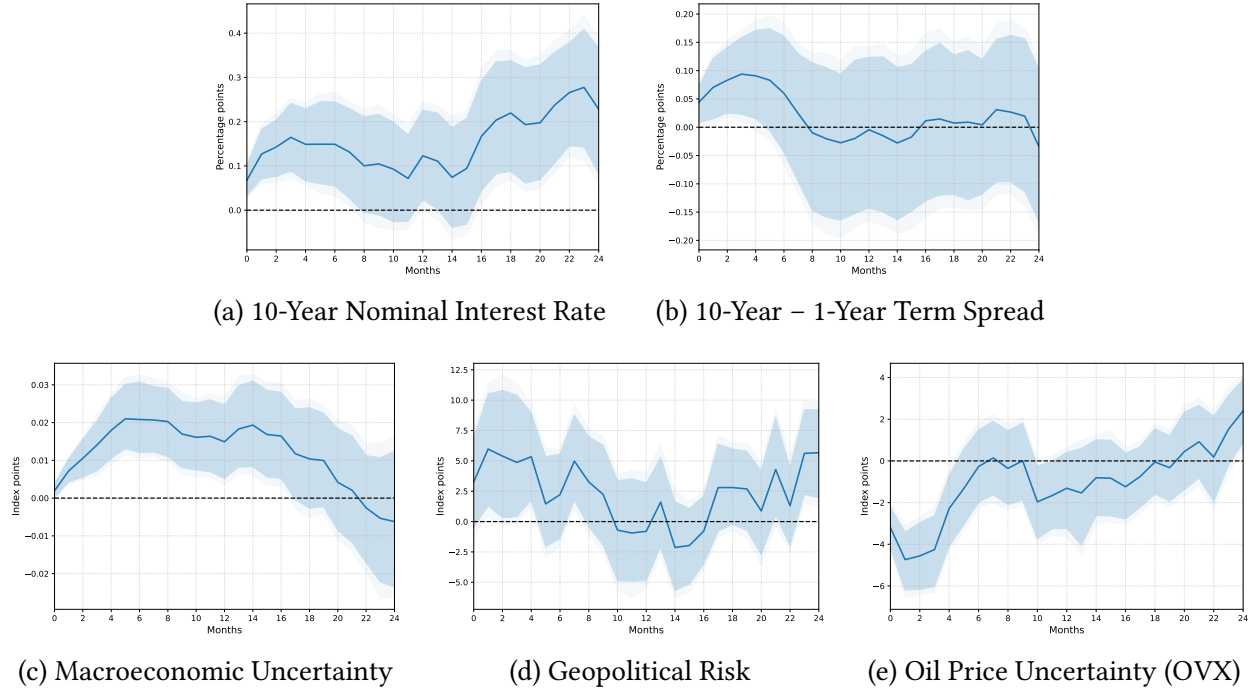
In contrast, wind drops slightly in the first 6 months but the variation in the point estimates points to noise as opposed to a proper steady decline. Hydro in turn actually increases in the first 6 months, peaking at 1.75 percentage points but this increase is short-lived, and it actually also declines 1 year after the increase in oil prices, by 0.89 percentage points, before rebounding.

4.2.3 The Drivers of Solar’s Decline

The results for solar take-up reflect three compounding mechanisms. First, solar power is intermittent — it generates only when irradiance is sufficient — and therefore requires fast-reacting dispatchable backup capacity, primarily natural gas peaker plants, to balance the grid during periods of low generation (Verdolini, Vona, and Popp 2018; Stöckl and Zerrahn 2023). When oil shocks simultaneously disrupt natural gas supply and raise gas prices, the effective cost of deploying complementary solar capacity rises, dampening new investment. Second, the inflationary nature of the shock triggers an endogenous tightening of monetary policy (Bernanke, Gertler, and Watson 1997), causing credit conditions to tighten — an environment that does not support capital-intensive activities such as solar installation (Bistline, Mehrotra, and Wolfram 2023; Aghion et al. 2024; Ameli et al. 2021; Chen and Lin 2024). Third, and separately, macroeconomic and geopolitical uncertainty rise sharply following the shock. Unlike the credit channel, elevated uncertainty is a distinct deterrent to long-lived, high fixed-cost investments, by raising the option value of waiting (Bloom 2009).

Aggregate local projections confirm the latter two channels. Figure 13 shows that the 10-year nominal interest rate, obtained from FRED, rises sharply on impact, reaching 0.14 percentage points after 2 months. The term spread initially rises as long-term rates react before short-term rates catch up with monetary policy tightening, after which the spread narrows. Macroeconomic uncertainty (Jurado, Ludvigson, and Ng 2015) spikes on impact and remains elevated, and geopolitical risk (Caldara and Iacoviello 2022) rises persistently — all consistent with the decline in solar generation documented in panel *b*) of figure 12. In contrast, oil price uncertainty does not rise: the CBOE crude oil volatility index (OVX), obtained from FRED, falls in the first months after the shock (panel *e*)). The pattern is similar for the structural supply shocks of Baumeister and Hamil-

Figure 13: Financial Barriers to Clean Energy Adoption Following an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable in each panel is the $h + 1$ long-difference in the respective series. Macroeconomic uncertainty is the 3-month-ahead index of Jurado, Ludvigson, and Ng (2015); geopolitical risk is the index of Caldara and Iacoviello (2022); oil price uncertainty is the CBOE crude oil volatility index (OVX), the 30-day volatility of crude oil prices implied by option prices (monthly average, from FRED). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024, except panel (e), which starts in June 2008 because the OVX is only available from June 2007; outcome data extends to December 2025.

ton (2019), so it is not particular to the identification around OPEC announcements (figure A.4). The uncertainty that deters investment is therefore macroeconomic and geopolitical, rather than specific to the oil market.

5 Conclusion

This paper documents the global environmental consequences of unexpected increases in crude oil prices following supply disruptions. A 10% increase in oil prices reduces atmospheric CO_2e concentration by 0.10 parts per million relative to trend. This decline is equivalent to 3.24% of the average annual increase in the atmospheric concentration of CO_2e over the 21st century.

Notwithstanding, the underlying dynamics of individual greenhouse gases reveal important countervailing forces. At the fossil fuel level, the response is heterogeneous — retail fuels fall first, while commercial fuels such as jet kerosene and bunker fuel fall later, and jet kerosene does

not recover within the 2-year horizon — but the decline is general. Natural gas consumption falls, reflecting supply disruption rather than demand substitution. On the contrary, coal exhibits a bifurcated response: steam and coking coal production decline alongside economic activity, while lignite — the cheapest and most CO_2 -intensive grade — rises on impact. Bioenergy generation rises. Consistent with this, the atmospheric concentration of N_2O , a potent and long-lived greenhouse gas, rises persistently.

The clean-energy transition does not speed up as a result of oil price increases. ICE registrations fall persistently and remain depressed for 2 years, my horizon of study. Similarly, non-ICE vehicle demand declines initially but rebounds faster, growing temporarily above trend. Notwithstanding, this does not represent an acceleration in alternative vehicle take-up, with the percentage of non-ICE vehicle registrations remaining statistically indistinguishable from its trend.

In the power sector, the story is similar. Total electricity generation falls by roughly the same amount as economic activity. Solar generation and installed capacity contract, driven by the combination of monetary-policy tightening that raises the cost of capital, rising geopolitical uncertainty, and grid-level complementarities between intermittent solar and dispatchable natural gas — whose supply is itself disrupted by the shock.

As a result, the emissions efficiency improvement observable in the very short term — emissions per unit of industrial production decline within two quarters following the shock — is a short-lived intensive-margin response to higher energy costs, not a structural decoupling of output from emissions. It resembles the cyclical covariation between activity and carbon documented by Heutel (2012) and Doda (2014) more closely than the permanent technological decoupling achievable through directed energy innovation (Hassler, Krusell, and Olovsson 2021; Känzig and Williamson 2024).

These findings carry implications for how oil market disruptions should be interpreted through a climate lens. An oil price shock delivers a partial and distorted carbon price signal: it compresses activity in oil-intensive transport but simultaneously undermines capital-intensive renewable investment and stimulates low-grade combustion alternatives at the dispatchable margin.

Focusing on the green transition in the long run, future work should study how uncertainty in oil and gas production and the subsequent volatility in prices promote the transition to green energy. This mechanism is commonly alluded to in discussions surrounding oil disruptions, but is not observed in my short-run analysis, in part because other sources of uncertainty and financial constraints overwhelmingly affect the installation and development of clean energy alternatives.

References

Acemoglu, Daron, Philippe Aghion, Lint Barrage, and David Hémous. 2023. *Climate change, directed innovation, and energy transition: the long-run consequences of the shale gas revolution*.

- Working Paper 31657. National Bureau of Economic Research, September. <https://doi.org/10.3386/w31657>.
- Aghion, Philippe, Antonin Bergeaud, Maarten De Ridder, and John Van Reenen. 2024. *Lost in transition: financial barriers to green growth*. SSRN Scholarly Paper 4771433. Rochester, NY: Social Science Research Network, March 25, 2024. <https://doi.org/10.2139/ssrn.4771433>.
- Almuzara, Martín, and Víctor Sancibrián. 2026. *Micro responses to macro shocks*. Staff Report 1090. Federal Reserve Bank of New York, August.
- Ameli, Nadia, Olivier Dessens, Matthew Winning, Jennifer Cronin, Hugues Chenet, Paul Drummond, Alvaro Calzadilla, Gabriel Anandarajah, and Michael Grubb. 2021. Higher cost of finance exacerbates a climate investment trap in developing economies. *Nature Communications* 12, no. 1 (June 30, 2021): 4046. ISSN: 2041-1723. <https://doi.org/10.1038/s41467-021-24305-3>.
- Andrew, Robbie. 2026. Collected vehicle registration data. Collected vehicle registration data, June. Accessed June 8, 2026. <https://robbieandrew.github.io/carsales>.
- Baumeister, Christiane, and James D. Hamilton. 2019. Structural interpretation of vector autoregressions with incomplete identification: revisiting the role of oil supply and demand shocks. *American Economic Review* 109, no. 5 (May 1, 2019): 1873–1910. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20151569>.
- . 2025. *Uncovering disaggregated oil market dynamics: a full-information approach to granular instrumental variables*. April 26, 2025.
- Baumeister, Christiane, Florian Huber, Thomas K. Lee, and Francesco Ravazzolo. 2024. *Forecasting natural gas prices in real time*. Working Paper 33156. National Bureau of Economic Research, November. <https://doi.org/10.3386/w33156>.
- Bernanke, Ben S., Mark Gertler, and Mark W. Watson. 1997. Systematic monetary policy and the effects of oil price shocks, Brookings Papers on Economic Activity.
- Bistline, John E. T., Neil R. Mehrotra, and Catherine Wolfram. 2023. Economics implications of the climate provisions of the inflation reduction act. *Brookings Papers on Economic Activity* 2023 (2): 77–157.
- Bloom, Nicholas. 2009. The impact of uncertainty shocks. *Econometrica* 77 (3): 623–685. ISSN: 1468-0262. <https://doi.org/10.3982/ECTA6248>.
- Caldara, Dario, and Matteo Iacoviello. 2022. Measuring geopolitical risk. *American Economic Review* 112, no. 4 (April): 1194–1225. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20191823>.
- Casey, Gregory. 2024. Energy efficiency and directed technical change: implications for climate change mitigation. *The Review of Economic Studies* 91, no. 1 (January 1, 2024): 192–228. ISSN: 0034-6527. <https://doi.org/10.1093/restud/rdad001>.

- Chen, Shiu-Sheng, and Tzu-Yu Lin. 2024. Monetary policy and renewable energy production. *Energy Economics* 132 (April 1, 2024): 107495. ISSN: 0140-9883. <https://doi.org/10.1016/j.eneco.2024.107495>.
- Chodorow-Reich, Gabriel, Plamen T. Nenov, and Alp Simsek. 2021. Stock market wealth and the real economy: a local labor market approach. *American Economic Review* 111, no. 5 (May): 1613–1657. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20200208>.
- Climate Watch. 2021. Historical GHG emissions. Accessed February 17, 2025. https://www.climatewatchdata.org/ghg-emissions?apcid=000000000000000000000000&breakBy=sector&calculation=PERCENTAGE_CHANGE&chartType=line&end_year=2021§ors=&start_year=1990.
- Doda, Baran. 2014. Evidence on business cycles and CO2 emissions. *Journal of Macroeconomics* 40 (June 1, 2014): 214–227. ISSN: 0164-0704. <https://doi.org/10.1016/j.jmacro.2014.01.003>.
- Eggleston, Simon, Leandro Buendia, Kyoko Miwa, Todd Ngara, and Kiyoto Tanabe, eds. 2006. *2006 IPCC guidelines for national greenhouse gas inventories, prepared by the national greenhouse gas inventories programme*. Japan: Institute for Global Environmental Strategies (IGES) for the IPCC.
- Etminan, M., G. Myhre, E. J. Highwood, and K. P. Shine. 2016. Radiative forcing of carbon dioxide, methane, and nitrous oxide: a significant revision of the methane radiative forcing. *Geophysical Research Letters* 43 (24): 12, 614–12, 623. ISSN: 1944-8007. <https://doi.org/10.1002/2016GL071930>.
- Government of Canada. 2024. Canada’s greenhouse gas quantification requirements (2024). Last Modified: 2024-04-03, April 3, 2024. Navigation page;guidance. Accessed August 22, 2026. <https://www.canada.ca/en/environment-climate-change/services/climate-change/greenhouse-gas-emissions/facility-reporting/reporting/quantification-requirements/canada-greenhouse-gas-reporting-program-2024.html>.
- Hamilton, James D. 1983. Oil and the macroeconomy since world war II. *Journal of Political Economy* (Chicago, IL) 91, no. 2 (April): 228–248.
- Harstad, Bård, and Katinka Holtmark. 2026. The gas trap: outcompeting coal versus renewables. (Forthcoming), *Journal of Political Economy*.
- Hassler, John, Per Krusell, and Conny Olovsson. 2021. Directed technical change as a response to natural resource scarcity. *Journal of Political Economy* 129, no. 11 (November 1, 2021): 3039–3072. ISSN: 0022-3808, 1537-534X. <https://doi.org/10.1086/715849>.
- Heutel, Garth. 2012. How should environmental policy respond to business cycles? optimal policy under persistent productivity shocks. *Review of Economic Dynamics* 15, no. 2 (April 1, 2012): 244–264. ISSN: 1094-2025. <https://doi.org/10.1016/j.red.2011.05.002>.

- Intergovernmental Panel on Climate Change (IPCC). 2023a. Changing state of the climate system. In *Climate change 2021 – the physical science basis: working group i contribution to the sixth assessment report of the intergovernmental panel on climate change*, 287–422. Cambridge University Press.
- . 2023b. The earth’s energy budget, climate feedbacks and climate sensitivity. In *Climate change 2021 – the physical science basis: working group i contribution to the sixth assessment report of the intergovernmental panel on climate change*, 923–1054. Cambridge University Press.
- International Energy Agency. 2023. How is oil used globally? Accessed June 13, 2026. <https://www.iea.org/world/oil>.
- Jordà, Òscar. 2005. Estimation and inference of impulse responses by local projections. *American Economic Review* 95, no. 1 (February 1, 2005): 161–182. ISSN: 0002-8282. <https://doi.org/10.1257/0002828053828518>.
- Jurado, Kyle, Sydney C. Ludvigson, and Serena Ng. 2015. Measuring uncertainty. *American Economic Review* 105, no. 3 (March 1, 2015): 1177–1216. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20131193>.
- Känzig, Diego R. 2021. The macroeconomic effects of oil supply news: evidence from OPEC announcements. *American Economic Review* 111 (4).
- Känzig, Diego R., Maximilian Konradt, Lixing Wang, and Donghai Zhang. 2025. *Green business cycles*. Working Paper 34041. National Bureau of Economic Research, July. <https://doi.org/10.3386/w34041>.
- Känzig, Diego R., and Charles Williamson. 2024. Unraveling the drivers of energy-saving technical change. *SSRN Electronic Journal*, ISSN: 1556-5068. <https://doi.org/10.2139/ssrn.4973880>.
- Khan, Hashmat, Konstantinos Metaxoglou, Christopher R. Knittel, and Maya Papineau. 2019. Carbon emissions and business cycles. *Journal of Macroeconomics* 60 (June 1, 2019): 1–19. ISSN: 0164-0704. <https://doi.org/10.1016/j.jmacro.2019.01.005>.
- Kilian, Lutz. 2009. Not all oil price shocks are alike: disentangling demand and supply shocks in the crude oil market. *The American Economic Review* 99 (3): 1053–1069. ISSN: 0002-8282.
- Kilian, Lutz, and Xiaoqing Zhou. 2024. Heterogeneity in the pass-through from oil to gasoline prices: a new instrument for estimating the price elasticity of gasoline demand. *Journal of Public Economics* 232 (April): 105099. ISSN: 00472727. <https://doi.org/10.1016/j.jpubeco.2024.105099>.
- Lan, Xin, Pieter Tans, and K.W. Thoning. 2026. *Trends in globally-averaged CO2 determined from NOAA global monitoring laboratory measurements*, May 5, 2026. <https://doi.org/10.15138/9N0H-ZH07>.

- Ramey, Valerie A. 2019. Ten years after the financial crisis: what have we learned from the renaissance in fiscal research? *Journal of Economic Perspectives* 33, no. 2 (May): 89–114. ISSN: 0895-3309. <https://doi.org/10.1257/jep.33.2.89>.
- Stöckl, Fabian, and Alexander Zerrahn. 2023. Substituting clean for dirty energy: a bottom-up analysis. *Journal of the Association of Environmental and Resource Economists* 10, no. 3 (May): 819–863. ISSN: 2333-5955. <https://doi.org/10.1086/722612>.
- U.S. Energy Information Administration. 2013. Drilling often results in both oil and natural gas production - u.s. energy information administration (EIA), October 29, 2013. Accessed May 30, 2026. <https://www.eia.gov/todayinenergy/detail.php?id=13571>.
- Verdolini, Elena, Francesco Vona, and David Popp. 2018. Bridging the gap: do fast-reacting fossil technologies facilitate renewable energy diffusion? *Energy Policy* 116 (May 1, 2018): 242–256. ISSN: 0301-4215. <https://doi.org/10.1016/j.enpol.2018.01.058>.